

**SIMATS ENGINEERING**  
**Department of Computer Science and Engineering**  
**ASSESSMENT TOOL 2 – COMPARATIVE ANALYSIS**

|                         |                                                                                                                                                                                                                                                    |                      |                       |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-----------------------|
| <b>Course Code:</b>     | CSA12                                                                                                                                                                                                                                              | <b>Course Title:</b> | Computer Architecture |
| <b>CO Assessed:</b>     | CO4 – Precision (BL4)                                                                                                                                                                                                                              | <b>Assessment:</b>   | Comparative Analysis  |
| <b>Weightage:</b>       | 30%                                                                                                                                                                                                                                                | <b>Total Marks:</b>  | 100                   |
| <b>CO4 Statement:</b>   | Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics. |                      |                       |
| <b>Student Name:</b>    | SIVARANGJENII B                                                                                                                                                                                                                                    | <b>Reg. No.:</b>     | 192521375             |
| <b>Section / Batch:</b> | B.TECH IT                                                                                                                                                                                                                                          | <b>Date:</b>         | 24.8.26               |

**Instructions to Students**

- Identify the relevant concepts of L1, L2, and L3 cache, SRAM, DRAM, NAND Flash, MRAM, RRAM, virtual memory, paging, latency, bandwidth, and throughput.
- Analyze the memory requirements of different computer systems by considering cache levels, memory technologies, access latency, bandwidth, throughput, capacity, and energy consumption.
- Evaluate the performance characteristics of different memory technologies and determine their suitability for cache memory, main memory, secondary storage, and future hierarchical memory systems.
- Compute relevant performance measures such as access time, latency, bandwidth, throughput, hit/miss rate, average memory access time (AMAT), speedup, and energy efficiency wherever applicable.
- Interpret the results by assessing memory bottlenecks, performance improvements, technology trade-offs, and the suitability of each memory technology or memory hierarchy for a given application.

**QUESTIONS**

1. Evaluate L1, L2, and L3 cache levels based on access latency, bandwidth, and throughput. Analyze how each cache level influences processor performance and memory access efficiency.
2. Assess the performance characteristics of SRAM and DRAM with respect to access latency, bandwidth, and throughput. Justify the use of SRAM in cache memory and DRAM in main memory based on their respective characteristics.
3. Examine virtual memory, paging, and cache memory mechanisms in terms of access latency, memory utilization, and system throughput. Analyze how these mechanisms contribute to efficient memory management and overall system performance.
4. Analyze the performance characteristics of NAND Flash and DRAM with respect to access latency, bandwidth, and throughput. Evaluate their suitability for different levels of a hierarchical memory system and justify their applications.
5. Evaluate MRAM and RRAM technologies based on access time, throughput, energy efficiency, and endurance. Analyze their potential applications in future hierarchical memory systems and assess

their performance advantages and limitations.

### Code of Conduct

I \_\_\_\_\_ B.SIVARANGJENII(192521375) \_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: \_\_\_\_\_

### RUBRICS FOR CALCULATION

| Criteria                | Weightage(%) | Level 4 – Exemplary (5 Marks)                                                 | Level 3 – Proficient (4 Marks)                  | Level 2 – Developing (2–3 Marks)           | Level 1 – Beginning (0–1 Mark)           |
|-------------------------|--------------|-------------------------------------------------------------------------------|-------------------------------------------------|--------------------------------------------|------------------------------------------|
| Scope of Items          | 20%          | Selects highly relevant items with clear scope and justification              | Selects appropriate items with minor gaps       | Selection is limited or partially relevant | Items are unclear or not relevant        |
| Comparison Depth        | 25%          | Provides detailed and comprehensive comparison across multiple aspects        | Provides adequate comparison with some depth    | Limited comparison with few aspects        | Very minimal or no comparison            |
| Use of Evidence         | 20%          | Supports comparison with accurate data, examples, or references               | Uses some supporting evidence with minor gaps   | Limited or weak supporting evidence        | No supporting evidence provided          |
| Critical Thinking       | 20%          | Demonstrates strong critical thinking with clear interpretation and reasoning | Shows reasonable interpretation with minor gaps | Basic interpretation; lacks depth          | No meaningful interpretation             |
| Clarity of Presentation | 15%          | Well-organized, clear, and logically structured presentation                  | Generally clear with minor issues               | Somewhat unclear or poorly structured      | Disorganized and difficult to understand |

### Marks Evaluation:

| Q. No | Scope of Items (20) | Comparison Depth (25) | Use of Evidence (20) | Critical Thinking (20) | Clarity of Presentation (15) | Total (out of 100) |
|-------|---------------------|-----------------------|----------------------|------------------------|------------------------------|--------------------|
| 1     | / 4                 | / 4                   | / 4                  | / 4                    | / 4                          |                    |
| 2     | / 4                 | / 4                   | / 4                  | / 4                    | / 4                          |                    |

|   |      |      |      |      |      |       |
|---|------|------|------|------|------|-------|
| 3 | / 4  | / 4  | / 4  | / 4  | / 4  |       |
| 4 | / 4  | / 4  | / 4  | / 4  | / 4  |       |
| 5 | / 4  | / 4  | / 4  | / 4  | / 4  |       |
|   |      |      |      |      |      |       |
|   | / 20 | / 25 | / 20 | / 20 | / 15 | / 100 |

**Course Coordinator Faculty Incharge**

## 1.Introduction

Cache memory is a high-speed memory system located between the CPU and main memory. Modern processors generally use three cache levels—L1, L2, and L3—to reduce memory access time and improve processor performance. Each level differs in capacity, access latency, bandwidth, and throughput. These differences create a hierarchy that balances speed, storage capacity, and cost.

### Comparison of L1, L2 and L3 Cache

| Characteristic | L1 Cache                          | L2 Cache          | L3 Cache                   |
|----------------|-----------------------------------|-------------------|----------------------------|
| Size           | Smallest                          | Larger than L1    | Largest                    |
| Access Latency | Very low                          | Low               | Higher than L1 and L2      |
| Bandwidth      | Very high                         | High              | High                       |
| Throughput     | Very high                         | High              | High                       |
| Location       | Closest to CPU core               | Close to CPU core | Usually shared among cores |
| Main Purpose   | Frequently used data/instructions | Backup for L1     | Larger shared cache        |

The exact latency and bandwidth values vary between processor architectures, so these represent the general performance relationship.

### Access Latency

L1 cache has the lowest access latency because it is very small and located closest to the processor core. This allows frequently required instructions and data to be accessed almost immediately.

L2 cache has slightly higher latency because it is larger than L1. However, it is still considerably faster than accessing main memory. When data is not available in L1, L2 can provide it without requiring a DRAM access.

L3 cache has higher latency than L1 and L2, but its larger capacity allows it to store more useful data. Since L3 is commonly shared between processor cores, it can also support data access across multiple cores.

**Therefore, the general relationship is:**

L1 → lowest latency → L2 → higher latency → L3 → higher latency

### **Bandwidth and Throughput**

L1 generally provides the highest effective bandwidth because it is closest to the CPU and is designed for very rapid data transfers. High bandwidth allows the processor to receive required data quickly.

L2 provides high bandwidth while offering greater capacity than L1. It acts as an intermediate level between the very fast L1 and the larger L3 cache.

L3 provides high bandwidth and large capacity, although its access performance is generally lower than L1 and L2. Its major advantage is reducing the number of accesses to slower main memory.

High cache throughput means that the processor can complete more operations without waiting for data from DRAM.

### **Influence on Processor Performance**

The cache hierarchy improves processor performance mainly by increasing the probability of a cache hit.

**For example:**

L1 hit → Very fast data access

L1 miss + L2 hit → Data obtained without accessing DRAM

L1/L2 miss + L3 hit → DRAM access avoided

L1/L2/L3 miss → Main memory access required

Therefore, effective cache utilization reduces CPU waiting time and increases overall

processor throughput.

### **Memory Access Efficiency**

The three cache levels reduce the average memory access time of the processor. Frequently accessed data tends to remain in the cache, allowing the CPU to access it much faster than if it had to retrieve it from DRAM.

#### **The basic relationship can be represented as:**

Average Memory Access Time = Cache Access Time + Miss Rate  $\times$  Miss Penalty

A high cache-hit rate results in fewer accesses to slower memory and therefore improves memory access efficiency.

### **Performance Trade-Off**

The fastest cache is not necessarily the largest cache. L1 provides maximum speed but limited capacity, while L2 provides a balance between speed and capacity. L3 provides much greater capacity and is useful for multicore processors but has higher latency.

Thus, increasing cache capacity involves a trade-off between speed, capacity, hardware cost, and access latency. The processor achieves high performance by using all three levels together rather than relying on only one cache.

### **Real-World Application**

In applications such as gaming, video processing, databases, scientific computing, and artificial intelligence, the processor frequently accesses the same or nearby data. If this data remains in L1 or L2 cache, repeated accesses can be performed very quickly.

For example, when a processor repeatedly uses the same variables in a loop, keeping those variables in cache prevents repeated slow accesses to main memory. This improves execution speed and processor utilization.

### **Conclusion**

L1, L2, and L3 caches form a hierarchical system that balances latency, bandwidth, throughput, capacity, and cost. L1 provides the fastest access, L2 provides a balance

between speed and capacity, and L3 provides larger shared storage while reducing accesses to main memory.

Overall, the combined cache hierarchy reduces average memory access time, minimizes processor waiting, improves memory access efficiency, and increases overall processor performance.

## **2 Assess the Performance Characteristics of SRAM and DRAM**

### **Introduction**

SRAM (Static Random-Access Memory) and DRAM (Dynamic Random-Access Memory) are two important types of volatile semiconductor memory. Both are used in computer systems, but they have different characteristics in terms of access latency, bandwidth, throughput, capacity, cost, and power consumption. These differences determine why SRAM is mainly used for cache memory while DRAM is used for main memory.

### **Comparison of SRAM and DRAM**

| <b>Characteristic</b> | <b>SRAM</b>    | <b>DRAM</b>           |
|-----------------------|----------------|-----------------------|
| Access Latency        | Very low       | Higher than SRAM      |
| Bandwidth             | Very high      | High                  |
| Throughput            | Very high      | High                  |
| Storage Density       | Lower          | Higher                |
| Cost per Bit          | High           | Lower                 |
| Refresh Requirement   | Not required   | Required periodically |
| Power Consumption     | Higher per bit | Lower per bit         |
| Typical Application   | Cache memory   | Main memory           |

### **Access Latency**

SRAM has lower access latency because it stores each bit using a flip-flop circuit and does not require a refresh operation. Its simple and fast access mechanism allows the processor to retrieve data quickly.

DRAM has higher latency because it stores each bit using a capacitor that gradually loses its charge. The memory must therefore be refreshed periodically, adding additional operations to memory access.

**The general relationship is:**

SRAM → Lower latency → Faster access

DRAM → Higher latency → Slower access

This makes SRAM more suitable when the processor requires extremely fast data access.

**Bandwidth and Throughput**

SRAM can provide very high bandwidth and throughput because of its fast access characteristics. This allows the CPU to obtain frequently used instructions and data quickly.

DRAM also provides high bandwidth and is designed to transfer large amounts of data efficiently. However, its access latency is greater than SRAM.

In a processor, high memory throughput is important because slow memory access can cause the CPU to wait for data. SRAM minimizes this waiting time when data is available in the cache.

**Memory Density and Cost**

One major advantage of DRAM is its high storage density. A large amount of data can be stored in a relatively small physical area. It is also significantly cheaper per bit than SRAM.

SRAM requires more transistors to store each bit, making it more expensive and requiring more physical space.

**Therefore:**

SRAM → Faster but expensive and lower density



DRAM → Slower but cheaper and higher density

This difference is one of the main reasons why large computer systems cannot practically use SRAM for all their main memory.

### **Why SRAM is Used in Cache Memory?**

Cache memory needs extremely fast access because it is directly involved in supplying data to the processor.

#### **If SRAM is used as cache:**

CPU → SRAM Cache → Fast data access → Less CPU waiting

The small size of cache makes SRAM's high cost manageable. Its low latency and high throughput are more important than its high cost.

For example, frequently used instructions and data can be stored in SRAM-based cache so that the CPU does not repeatedly access slower DRAM.

### **Why DRAM is Used in Main Memory?**

Main memory needs to provide a large storage capacity at reasonable cost. DRAM is well suited for this requirement because it has high density and a relatively low cost per bit.

For example, a computer may require several gigabytes of main memory. Implementing this entire capacity using SRAM would be extremely expensive and would require considerably more physical space.

Therefore, DRAM provides a practical balance between capacity, cost, bandwidth, and performance.

### **Performance Trade-Off**

The choice between SRAM and DRAM involves a clear trade-off.

#### **SRAM:**

Low latency + high throughput + high cost + low density

#### **DRAM:**

Higher latency + high bandwidth + low cost + high density

Therefore, the fastest memory is not always the most practical memory. Computer

architecture combines both technologies to obtain an efficient memory hierarchy.

### **Impact on Overall System Performance**

Using SRAM for cache reduces the number of accesses to DRAM. When frequently used data is found in the cache, the processor can access it quickly.

If the data is not found in SRAM cache, the processor must access DRAM, which takes longer.

#### **Therefore:**

Higher SRAM cache hit rate → Fewer DRAM accesses → Lower average memory latency → Higher CPU performance

This combination allows the system to achieve both high speed and large memory capacity.

### **Conclusion**

SRAM and DRAM have different strengths. SRAM provides very low access latency, high bandwidth, and high throughput, making it ideal for processor cache. However, its high cost and low density prevent it from being used as large-capacity main memory.

DRAM provides high density, lower cost, and good bandwidth, making it suitable for main memory despite its higher access latency and refresh requirement.

Thus, using SRAM for cache and DRAM for main memory provides an effective balance between speed, capacity, cost, and overall system performance.

## **3. Examine Virtual Memory, Paging, and Cache Memory Mechanisms**

### **Introduction**

Virtual memory, paging, and cache memory are important memory-management mechanisms used in modern computer systems. Although all three improve memory utilization and system performance, they operate at different levels and solve different problems. Their effectiveness can be analyzed using access latency,

memory utilization, and system throughput.

### Comparison of Virtual Memory, Paging, and Cache Memory

| Characteristic     | Cache Memory              | Paging                                | Virtual Memory                                                      |
|--------------------|---------------------------|---------------------------------------|---------------------------------------------------------------------|
| Main Purpose       | Speed up data access      | Organize physical memory              | Extend available memory                                             |
| Access Latency     | Very low                  | Moderate                              | High when storage is accessed                                       |
| Memory Utilization | Improves through locality | Efficiently allocates physical memory | Allows programs larger than physical RAM                            |
| Throughput         | Increases CPU throughput  | Improves memory utilization           | Can increase multitasking but excessive swapping reduces throughput |
| Main Technology    | SRAM                      | Pages and frames                      | RAM + secondary storage                                             |
| Major Benefit      | Faster CPU access         | Efficient memory allocation           | Larger address space                                                |

### Cache Memory and Access Latency

Cache memory is located close to the processor and stores frequently accessed

instructions and data. Its primary purpose is to reduce memory access latency.

When the processor finds the required data in cache, it is called a cache hit. The data can be accessed much faster than data stored in main memory.

**For example:**

CPU → Cache hit → Fast access → Less CPU waiting

If the required data is not present, a cache miss occurs and the processor has to access a lower level of memory.

Therefore, a high cache-hit rate improves system performance and processor throughput.

**Paging and Memory Utilization**

Paging divides logical memory into fixed-size pages and physical memory into corresponding frames. A program's pages can be placed in different physical memory locations instead of requiring one continuous block.

This improves memory utilization because available physical memory can be allocated more flexibly.

For example, if a program requires several pages, those pages do not have to occupy consecutive physical memory locations.

However, paging introduces additional overhead because the operating system must maintain page tables and translate virtual addresses into physical addresses.

**Virtual Memory and Memory Capacity**

Virtual memory allows a program to use an address space that can be larger than the available physical RAM. It uses secondary storage as an extension of memory when required pages are not currently in RAM.

**This provides an important advantage:**

Program size > Available physical RAM → Virtual memory allows execution

However, secondary storage is significantly slower than RAM. When a required page is not in physical memory, a page fault occurs and the operating system must load the page from storage.

Therefore, virtual memory improves memory capacity and flexibility, but excessive page faults can significantly reduce system performance.

### **Access Latency Comparison**

The three mechanisms have very different latency characteristics.

Cache memory → Very low latency

Paging → Additional address-translation overhead

Virtual memory → Very high latency when page faults require secondary-storage access

Therefore, the system attempts to keep frequently accessed data in cache and actively used pages in RAM to avoid expensive accesses to secondary storage.

### **Effect on Memory Utilization**

Cache memory uses the principle of locality of reference. Frequently used data and instructions are kept close to the CPU, making memory accesses more efficient.

Paging allows physical memory to be divided into manageable units and prevents programs from requiring contiguous physical memory.

Virtual memory allows multiple programs to share physical memory and provides each process with its own virtual address space.

### **Thus:**

Cache → Efficient use of fast memory

Paging → Efficient organization of physical memory

Virtual memory → Efficient use of physical + secondary storage

### **Effect on System Throughput**

Cache memory generally increases throughput because the CPU spends less time waiting for data.

Paging can improve throughput by allowing memory to be allocated efficiently among multiple processes. However, excessive page-table operations and page faults can introduce overhead.

Virtual memory supports multitasking and allows more programs to remain active.

However, if the system continuously moves pages between RAM and storage, thrashing may occur. Thrashing causes CPU utilization and overall throughput to decrease significantly.

Therefore, efficient memory management requires keeping the number of page faults under control.

### Interaction Between the Mechanisms

These mechanisms work together rather than independently.

#### **A simplified memory-access path is:**

CPU → Cache → Main Memory → Virtual Memory/Storage

Frequently used data is kept in cache for very fast access. If it is not available in cache, the system accesses RAM. If the required page is not present in RAM, virtual memory mechanisms may require it to be retrieved from secondary storage.

This hierarchy ensures that the fastest available memory is used whenever possible.

#### **Critical Analysis**

Cache memory primarily solves the speed problem, while paging and virtual memory primarily solve memory-management and capacity problems.

Cache provides the lowest latency but has limited capacity. Paging improves the organization of physical memory but introduces address-translation overhead.

Virtual memory provides flexibility and allows programs to operate with limited RAM, but storage access during page faults is extremely slow.

Therefore, none of these mechanisms should be considered independently. Their combined operation determines the efficiency of the memory system.

#### **Real-World Example**

Consider a computer running a web browser, video player, and document editor simultaneously.

Cache memory keeps frequently used processor instructions and data close to the CPU. Paging allows the operating system to divide memory among different applications efficiently. Virtual memory allows less frequently used pages to be

moved out of RAM when physical memory becomes limited.

If the system has sufficient RAM and a high cache-hit rate, applications run smoothly. If RAM becomes heavily overloaded and frequent page faults occur, performance decreases because the system spends more time accessing secondary storage.

## **Conclusion**

Cache memory, paging, and virtual memory contribute to memory efficiency in different ways. Cache memory minimizes access latency and improves CPU throughput, paging improves physical memory organization, and virtual memory increases the effective memory capacity of the system.

Their combined use creates an efficient memory hierarchy. However, excessive cache misses, page faults, or virtual-memory activity can reduce performance. Therefore, effective memory management aims to maximize cache hits, use physical memory efficiently, and minimize costly storage accesses.

4 and 5

## **4. Analyze the Performance Characteristics of NAND Flash and DRAM**

### **Introduction**

NAND Flash and DRAM are two important memory technologies used at different levels of the computer memory hierarchy. DRAM is volatile, fast-access memory used mainly as main memory, whereas NAND Flash is non-volatile and is widely used for persistent storage such as SSDs and memory cards. Their differences in access latency, bandwidth, throughput, endurance, and storage capacity determine their suitability for different applications.

### **Comparison of NAND Flash and DRAM**

| <b>Characteristic</b> | <b>NAND Flash</b>                     | <b>DRAM</b>                                  |
|-----------------------|---------------------------------------|----------------------------------------------|
| Access Latency        | Higher                                | Much lower                                   |
| Bandwidth             | High, especially with parallel access | Very high                                    |
| Throughput            | High for sequential operations        | Very high for frequent read/write operations |
| Volatility            | Non-volatile                          | Volatile                                     |
| Storage Density       | Very high                             | Lower than NAND Flash                        |
| Cost per Bit          | Low                                   | Higher                                       |
| Write Endurance       | Limited                               | Very high                                    |
| Refresh               | Not required                          | Required                                     |
| Main Application      | SSDs, memory cards, storage           | Main memory                                  |

### **Access Latency**

DRAM provides much lower access latency than NAND Flash. The processor can access data in DRAM quickly, making it suitable for active programs and frequently accessed data.

NAND Flash has higher latency because it stores data using floating-gate or charge-trap cells and performs operations at the page and block level. Its write and erase operations are particularly slower than DRAM operations.

### **Therefore:**

DRAM → Low latency → Suitable for active processing

NAND Flash → Higher latency → Suitable for persistent storage

### **Bandwidth and Throughput**

DRAM provides very high bandwidth and is designed to continuously supply data to



the processor. This makes it suitable for applications requiring frequent and random memory accesses.

NAND Flash can provide high throughput, particularly when multiple flash channels operate in parallel. Modern SSDs can therefore achieve high sequential read and write performance. However, NAND Flash generally has greater access latency and is less suitable for the processor's immediate memory requirements.

Thus, DRAM is optimized for low-latency memory access, while NAND Flash is optimized for high-capacity persistent storage.

### **Memory Capacity and Cost**

NAND Flash has a major advantage in storage density. Large amounts of data can be stored in a relatively small physical area, making it economical for high-capacity storage.

DRAM is more expensive per bit and is therefore not normally used to provide extremely large storage capacities.

### **This creates an important trade-off:**

NAND Flash → Large capacity + low cost per bit + non-volatile

DRAM → Fast access + high performance + volatile

### **Endurance and Data Retention**

DRAM can support extremely frequent read and write operations during normal operation. However, it is volatile, meaning its contents disappear when power is removed.

NAND Flash is non-volatile and retains information without power. This makes it ideal for permanent storage. However, NAND cells have a limited number of program/erase cycles, so techniques such as wear leveling and error correction are used in flash storage systems to improve reliability and lifespan.

### **Suitability in a Memory Hierarchy**

DRAM is positioned much closer to the processor because of its low access latency.

### **A simplified hierarchy is:**

CPU → Cache → DRAM → NAND Flash

### **As we move down the hierarchy:**

Capacity increases → Cost per bit decreases → Latency generally increases

Therefore, DRAM is suitable for main memory, while NAND Flash is suitable for secondary or persistent storage.

### **Performance Impact**

Using DRAM as main memory allows the CPU to access active programs and data quickly. If the processor had to obtain frequently used data directly from NAND Flash, the higher latency would significantly reduce performance.

NAND Flash reduces the need for traditional mechanical storage and provides fast, reliable, non-volatile storage. However, it should not replace DRAM completely because its access characteristics are not suitable for the processor's main working memory.

### **Real-World Applications**

#### **DRAM is commonly used in:**

- Main memory of computers
- Laptops
- Smartphones
- Servers
- Graphics-related memory systems

#### **NAND Flash is commonly used in:**

- SSDs
- USB flash drives
- Memory cards
- Smartphones
- Embedded storage systems

For example, when a program is running, its active instructions and data are primarily held in DRAM. When the program or its files need to be permanently stored, NAND Flash can retain that information even after the system is switched off.

### **Critical Evaluation**

DRAM offers superior latency and excellent throughput for active computing, but it is volatile and more expensive per bit. NAND Flash provides much higher density, lower cost per bit, and non-volatility, but its latency and write endurance make it unsuitable as a direct replacement for DRAM.

Therefore, their different characteristics are complementary rather than competing.

### **Conclusion**

NAND Flash and DRAM occupy different positions in the memory hierarchy because they solve different requirements. DRAM is optimized for fast, frequent access and is therefore suitable for main memory, while NAND Flash is optimized for high-density, persistent and economical storage.

The combination of both provides an effective balance between performance, capacity, cost, and data persistence.

## **5. Evaluate MRAM and RRAM Technologies**

### **Introduction**

MRAM (Magnetoresistive Random-Access Memory) and RRAM (Resistive Random-Access Memory) are emerging non-volatile memory technologies designed to overcome some limitations of traditional memory technologies. Both retain data without power and have the potential to occupy positions between conventional cache, main memory, and storage in future hierarchical memory systems.

Their suitability can be evaluated using access time, throughput, energy efficiency, endurance, density, and scalability.

## Comparison of MRAM and RRAM

| Characteristic    | MRAM             | RRAM                           |
|-------------------|------------------|--------------------------------|
| Access Time       | Very low         | Low                            |
| Throughput        | High             | Potentially high               |
| Energy Efficiency | High             | Very high potential            |
| Endurance         | Very high        | High, but technology-dependent |
| Volatility        | Non-volatile     | Non-volatile                   |
| Density           | High             | Potentially very high          |
| Data Retention    | Excellent        | Excellent                      |
| Main Challenge    | Cost and scaling | Variability and endurance      |
| Potential         | Cache/embedded   | Dense memory/storage           |
| Application       | memory           |                                |

### Access Time

MRAM provides fast read and write access because it uses magnetic states to represent information. Its combination of low latency and non-volatility makes it attractive for applications requiring fast persistent memory.

RRAM can also provide fast switching between different resistance states. Its simple cell structure provides strong potential for high-density memory and fast access.

However, actual performance varies depending on the specific MRAM or RRAM implementation.

### Throughput

MRAM can provide high throughput while supporting frequent read/write

operations. Its high endurance makes it suitable for applications where memory is accessed repeatedly.

RRAM also has the potential for high throughput, particularly when multiple memory cells can be accessed in parallel. Its resistance-based operation also makes it attractive for emerging computing architectures.

Therefore, both technologies have potential to reduce the performance gap between conventional RAM and persistent storage.

### **Energy Efficiency**

Energy efficiency is an important advantage of emerging non-volatile memories.

MRAM does not require continuous refresh to retain data. This can reduce standby power compared with volatile memories that require refresh operations.

RRAM also has strong potential for low-energy operation because data can be stored as resistance states without continuous power. This makes RRAM attractive for low-power devices and energy-efficient computing.

However, write energy can vary significantly depending on the technology and device design.

### **Endurance**

MRAM generally provides very high endurance, making it suitable for applications involving frequent writes.

RRAM can also provide high endurance, but endurance can be affected by material properties, switching mechanisms, and device variability.

Therefore, MRAM currently has an important advantage where extremely frequent and reliable write operations are required, while RRAM remains highly promising but continues to face engineering challenges.

### **Density and Scalability**

RRAM has strong potential for high density because of its relatively simple memory-cell structure. This makes it attractive for future high-capacity memory systems.

MRAM also offers good density, but scaling the technology while maintaining

performance, reliability, and manufacturing cost remains an important challenge.

**Therefore:**

MRAM → Strong in speed, endurance and non-volatility

RRAM → Strong potential in density, scalability and energy efficiency

**Applications in Future Memory Hierarchy**

**MRAM could potentially be used as:**

CPU/embedded cache → Fast non-volatile memory → Main-memory applications

Its low latency and high endurance make it particularly attractive for embedded systems and persistent cache applications.

**RRAM could potentially be used as:**

Dense memory → Storage-class memory → In-memory computing

RRAM is especially interesting for in-memory and neuromorphic computing, where the physical resistance states of memory cells can potentially be used to perform computation as well as store information.

**Performance Advantages**

**The major potential advantages of MRAM include:**

- Low access latency
- High endurance
- Non-volatility
- Good energy efficiency
- Fast read/write operations

**The major potential advantages of RRAM include:**

- High density potential
- Low power consumption
- Non-volatility
- Fast switching
- Suitability for emerging computing architectures

**Limitations and Challenges**

MRAM still faces challenges related to manufacturing complexity, cost, density, and scaling.

RRAM faces challenges such as device-to-device variability, endurance limitations in some implementations, switching reliability, and manufacturing consistency.

These limitations mean that neither technology has completely replaced SRAM, DRAM, or NAND Flash.

### **Critical Evaluation**

MRAM and RRAM should not simply be viewed as replacements for existing memory. Their greatest value may come from creating new intermediate levels in the memory hierarchy.

MRAM is particularly attractive when speed, endurance, and non-volatility are important. RRAM is particularly promising when high density, low energy consumption, and computational functionality are required.

Thus, the best technology depends on the application's requirements rather than one technology being universally superior.

### **Future Memory Hierarchy**

A possible future hierarchy could be:

CPU → SRAM/MRAM → DRAM → RRAM → NAND Flash

In this arrangement, MRAM could provide fast non-volatile memory closer to the processor, while RRAM could provide a larger and more energy-efficient intermediate layer before traditional NAND-based storage.

The exact hierarchy will depend on future improvements in manufacturing technology, cost, endurance, latency, and device reliability.

### **Conclusion**

MRAM and RRAM are promising technologies for future memory systems because they combine non-volatility with fast access and potentially high energy efficiency. MRAM has strong advantages in access speed and endurance, while RRAM offers significant potential for high density, low power consumption, and in-memory

computing.

Therefore, both technologies have the potential to complement or partially replace existing memory technologies in future hierarchical systems, but challenges involving cost, scalability, reliability, endurance, and manufacturing must be addressed before widespread adoption.